

Yanan Lyu (吕亚楠)

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The best way to predict the future is to invent it. - Alan Kay

Education

Jilin International Studies University

Changchun, Jilin

B.E. in Data Science and Big Data Technology

Sept 2022 - June 2026

- **Ranking:** 1/40 (**Top 2.5%**); **GPA:** 4.0/5.0; **English:** IELTS 6.0; CET-6 (468)
- **Awards:** National Scholarship for Encouragement (**Top 3%**) ×3, First Prize of Academic Scholarship ×2

Projects

Research on Sales Optimization Strategies for Unmanned Vending Machines Based on Big Data Analysis

Jilin, China

Chinese Collegiate Computing Competition

May 2024

- Develop a dynamic inventory prediction model to address overstocking and shortage issues.
- Combine user profiling and machine learning to enable personalized recommendations and optimize dynamic pricing strategies.
- Perform multi-dimensional analysis of sales data (spatial-temporal, user, product) to generate visual decision-support reports.
- Design a modular system architecture to support distributed processing and real-time early warning.

An Electronic Product Production Decision-Making Model Integrating Random Sampling and Cost-Benefit Analysis

Jilin, China

China Undergraduate Mathematical Contest in Modeling

December 2024

- Optimized decision thresholds using Monte Carlo simulation (100k runs) achieving ≥80% statistical power.
- Approximated NP-hard solutions via dynamic programming with 216-dimensional state matrix, improving efficiency by 40%.
- Implement real-time Bayesian updates via sliding windows, reducing parameter convergence time by 3× with <50 ms latency.
- Build a full life-cycle cost model integrating 12 cost factors, with <3% simulation error for million-unit production.

Publications

[1]. Hybrid Feature Extraction for Accurate Weather Image Classification

Yanan Lyu, Cheng Yao*, ZhiYan Wang, QiuYe Yu

The 7th International Conference on Artificial Intelligence Technologies and Applications, 2025. (ICAITA), Accepted.

[2]. CoupletEval: A Novel Benchmark for Assessing Chinese Linguistic Proficiency in Large Language Models

Yanan Lyu, QiuYe Yu*, ZhiYan Wang, Yan Liang, Fei-Fei Wang, Shu-Jun Li

The 7th International Conference on Artificial Intelligence Technologies and Applications, 2025. (ICAITA), Accepted.

[3]. Enhancing the Accuracy of Heart Disease Prediction: A Comparative Study of Random Forest and Gradient Boosting Models Based on Physiological Indicators

Di Wang, QiuYe Yu*, Yanan Lyu, ShuJun Li, Yan Liang, ZhiYan Wang*

The 7th International Conference on Artificial Intelligence Technologies and Applications, 2025. (ICAITA), Accepted.

Awards and Honors

2023.10 **National Scholarship for Encouragement**, Department of Education of Jilin Province

2024.10 **National Scholarship for Encouragement**, Department of Education of Jilin Province

2025.11 **National Scholarship for Encouragement**, Department of Education of Jilin Province

2024.09 & 2025.09 **The First Prize Scholarship**, Jilin International Studies University

2023.12 **Third Prize(National), 2023 MatherCup University Mathematical Modeling Challenge**, Chinese Society of Optimization, Overall Planning and Economic Mathematics

2024.05 **First Prize (Provincial), 2024 Chinese Collegiate Computing Competition**, Jilin Provincial Organizing Committee of Chinese Collegiate Computing Competition

2023.12 & 2014.12 **Second Prize (Provincial), China Undergraduate Mathematical Contest in Modeling (2 times)**, China Society for Industrial and Applied Mathematics

2024.07 **Bronze (Provincial), China International College Students Innovation Competition**, Jilin Provincial Organizing Committee